

MADU CHINEMEREM CLARENCE

Machine Learning Engineer | Lagos, Nigeria (Remote / Worldwide)

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PROFESSIONAL SUMMARY

Freelance Machine Learning Engineer with hands-on experience designing, training, and deploying end-to-end ML systems and distributed backend infrastructure. Proficient in Python, Rust, and cloud-native tooling (AWS, Docker, Kubernetes). Track record of delivering production-grade pipelines, open-source libraries, and real-world applied ML solutions. Deep interest in deep learning, MLOps, and systems design. Available for freelance engagements and open-source collaboration worldwide.

TECHNICAL SKILLS

Languages:	Python, Rust, SQL, Scala, C++, Julia, R, Java, MATLAB
ML / DL Frameworks:	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, NumPy, Pandas
MLOps & Infra:	Docker, Kubernetes, AWS (ECS Fargate, RDS, ElastiCache, ECR, Amazon MQ), GitHub Actions
Backend:	FastAPI, asyncpg, RabbitMQ, Redis, REST APIs, JWT Authentication
Data & Pipelines:	ETL pipeline design, time-series processing, IoT sensor data, async data streaming
Protocols:	BLE (Bluetooth Low Energy), TCP/IP, P2P networking, SHA-256 integrity verification
Tools:	Git, GitHub, Postman, VS Code, Streamlit, Linux/Bash
Concepts:	Deep Learning, NLP, Computer Vision, Anomaly Detection, Distributed Systems, Production ML Deployment, End-to-End ML, Open-Source Development

EXPERIENCE

Freelance Machine Learning Engineer

Self-Employed · Remote, Worldwide · 2023 – Present

- Architected and deployed Météo-Agri CI, a real-time agricultural weather monitoring system for Côte d'Ivoire using Streamlit and AWS, providing crop risk alerts and regional forecasts.
- Designed and published WaveShare, an open-source async P2P file transfer Python library using BLE peer discovery and direct TCP streaming with SHA-256 integrity verification; delivered with full CI pipeline, test suite, and OSS compliance files.
- Built a distributed financial backend comprising a Transaction Processing System (TPS) and Payment Gateway using FastAPI, RabbitMQ, asyncpg, Redis, and Docker Compose; designed for AWS ECS Fargate deployment with full integration test suite.
- Developed a headless Spotify release notification daemon using Spotipy and smtplib, structured for cron-based server deployment with zero browser or GUI dependencies.
- Implemented end-to-end ML pipelines across multiple client projects, covering data ingestion, feature engineering, model training, evaluation, and production deployment.

PROJECTS

Météo-Agri CI

Python · Streamlit · AWS · CI/CD · REST API

- Real-time agricultural weather monitoring platform serving farmers in Côte d'Ivoire.

- Integrated external weather APIs, implemented crop risk scoring logic, and deployed to a public Streamlit URL with a fully automated CI/CD pipeline.

WaveShare — Open-Source P2P File Transfer Library

Python · BLE (bleak) · TCP · SHA-256 · GitHub Actions

- Async peer-to-peer file transfer library enabling BLE-based device discovery and high-speed direct TCP streaming.
- Shipped as a full open-source repository including CLA, DCO, CI, and test suite.

Transaction Processing System + Payment Gateway

FastAPI · RabbitMQ · asyncpg · Redis · Docker · AWS

- Distributed financial backend with JWT authentication, polling-based async responses, and a background result consumer.
- Architected for AWS ECS Fargate, RDS, ElastiCache, Amazon MQ, and ECR deployment.

Spotify Release Notification System

Python · Spotipy · smtplib · Cron

- Headless backend daemon that monitors followed artists on Spotify and dispatches email alerts on new releases via scheduled cron jobs.

EDUCATION

B.Sc. Software Engineering (In Progress)

[Your University Name] · 100 Level · Expected Graduation: [Year]

- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Database Systems, Software Engineering Principles.

ADDITIONAL

- Open to remote freelance contracts, short-term projects, and open-source collaboration worldwide.
- Active GitHub contributor: github.com/Clarence6174
- ORCID: 0009-0009-6911-2442 | Medium: [@chinemeremmadu](https://medium.com/@chinemeremmadu)